Development of a Teacher-Focused Mobile Application for Colegio De Alicia Website

Members:

Amaya, Janssen Carl T.
Carbajal, John Albert V.
Galagar, John Mark H.
Lacorte, Daryl Bench P.
Mata, Beryl Paece C.
Santerna, Andrea Loren A.

Course: ELDROID12

Instructor: Wilson Gayo

School: University of Cebu – Main Campus

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ABSTRACT

This project focuses on **the design and development** of a mobile user interface for **teachers** of **Colegio De Alicia**. The goal of the application is to provide a structured and user-friendly interface where teachers can access important school-related modules such as schedules, grades, attendance monitoring, and profile information.

The application prototype was developed using Kotlin and Jetpack Compose in Android Studio. The system demonstrates modern mobile interface design principles using composable UI components and navigation architecture.

The project is limited to the **UI/UX** implementation of the application. Backend services, authentication systems, and database functionality are not included in the scope of this project. However, the prototype provides a foundation that can later be expanded into a fully functional teacher management system.

I. INTRODUCTION

1.1 Background of the Study

Mobile technology is widely used in education to help manage academic tasks more efficiently. Teachers often need quick access to information such as schedules, grades, and attendance records. However, many school processes are still done manually or through scattered systems, making it harder for teachers to organize their work.

To address this, the **Teacher-Side Application** was designed as a **mobile interface prototype** for teachers of Colegio De Alicia. The application provides an organized interface where teachers can navigate features such as dashboard, schedule, grades, attendance, and profile. The system was developed using **Android Studio**, **Kotlin**, and **Jetpack Compose** to demonstrate modern Android UI development.

1.2 Problem Statement

Many schools lack a dedicated mobile interface that allows teachers to easily manage academic information in one place. Because of this, teachers may find it difficult to quickly access schedules, record attendance, or manage grades.

This project aims to address the following questions:

1. How can a mobile interface be designed to help teachers manage academic tasks efficiently?
2. How can a simple and organized navigation system improve user experience for teachers?

1.3 Purpose of the Project

The purpose of this project is to design a mobile application user interface prototype for teachers of Colegio De Alicia. The system demonstrates how teachers can navigate different modules such as schedule, grades, attendance, and profile within a single application interface.

1.4 Significance of the Project

The mobile app is significant to:

Teachers: Provides a concept of a mobile platform that organizes important teaching-related tasks.

School Administration: Demonstrates how digital tools can support academic management.

Developers and Students: Serves as an example project for learning mobile UI development using modern Android tools.

II. OBJECTIVES

2.1 General Objective

To design and develop a mobile user interface prototype for teachers of Colegio De Alicia.

2.2 Specific Objectives

1. To create a clean and intuitive mobile interface for teacher users.
2. To implement UI components using Jetpack Compose.
3. To design a dashboard that serves as the central navigation point of the application.
4. To create modules for schedule viewing, grade management, attendance tracking, and profile information.
5. To demonstrate how a teacher management mobile application may function through a UI prototype.

III. SCOPE AND LIMITATIONS

3.1 Scope

The project includes:

- Mobile application user interface
- Dashboard interface for navigation
- Schedule viewing interface
- Grades interface layout
- Attendance interface layout
- Teacher profile interface
- Navigation system between screens

3.2 Limitations

The system does not include:

- Backend server integration
- Database connectivity
- Authentication systems

- Real-time data processing
- Student-side application

The application functions purely as a **UI/UX prototype**.

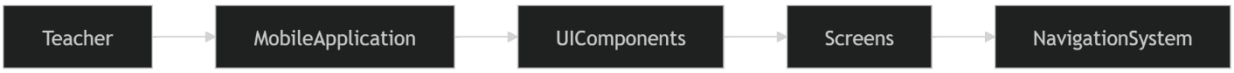
IV. DEVELOPMENT TOOLS AND TECHNOLOGIES

The application was developed using the following technologies:

TOOLS	DESCRIPTION
Android Studio	Main development environment
Kotlin	Programming language used to build the application
Jetpack Compose	Framework used to design the UI
Android Jetpack Navigation	Handles navigation between screens
GitHub	Used for version control and collaboration
Gradle	Used for project build management

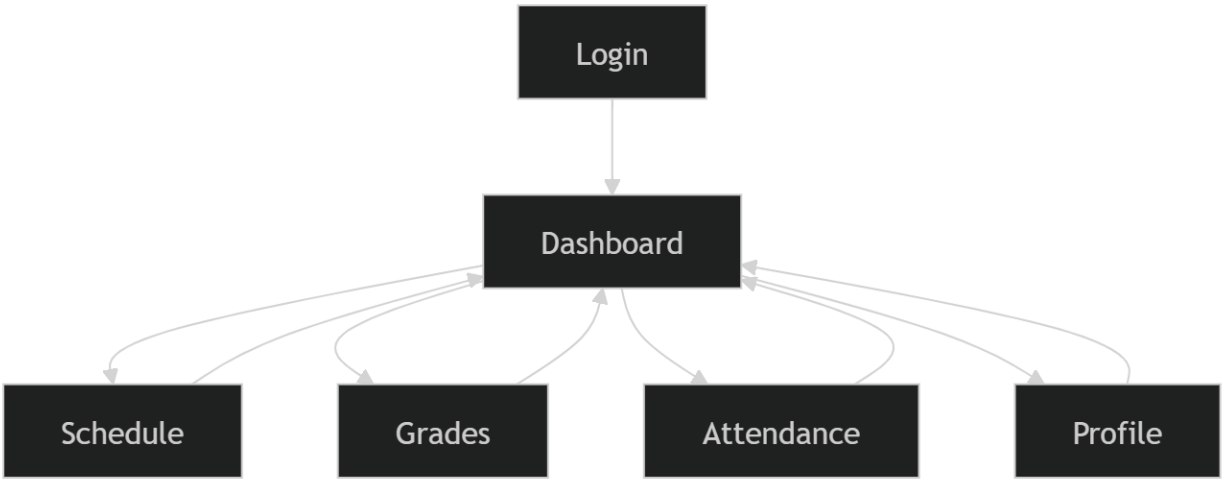
V. SYSTEM ARCHITECTURE

Since the project focuses only on the interface, the architecture follows a **UI-centered structure**.



The teacher interacts with the mobile application interface. The interface is composed of multiple screens connected through a navigation controller.

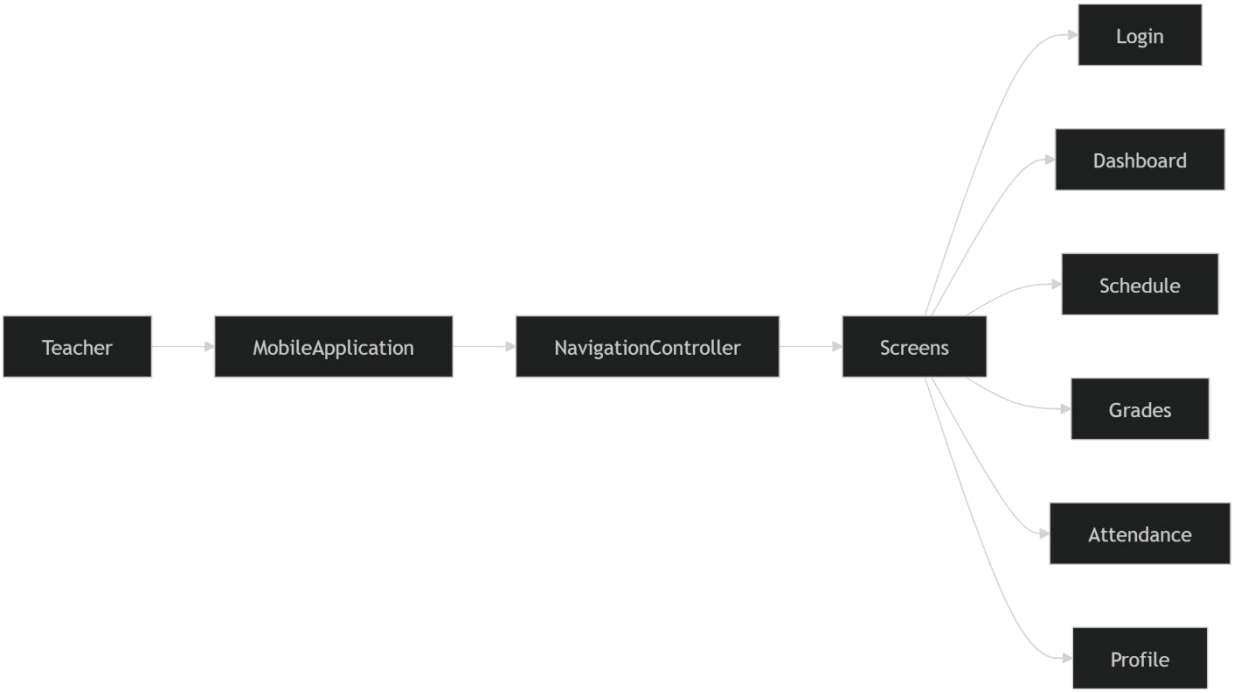
VI. NAVIGATION FLOW



Explanation:

1. The application starts with the Login Screen.
2. After login, the user is directed to the Dashboard Screen.
3. From the dashboard, teachers can access the other modules.

VII. UI ARCHITECTURE



VIII. APPLICATION MODULES

The application consists of several key modules that provide teachers with essential tools to manage their academic responsibilities. Each module focuses on a specific functionality and is designed to be intuitive and easy to navigate.

SCREEN	DESCRIPTION
Login	Acts as the **entry point** of the application. Allow teachers to access the system before navigating to the main interface.
Dashboard	The main screen of the application. Serves as the **navigation hub** where teachers can access the following modules**: Schedule, Grades, Attendance, Profile.**
Schedule	Displays the layout for **viewing teacher class schedules**. Help teachers monitor their daily teaching activities.
Grades	Provides the interface for **managing grading-related information**.
Attendance	Contains UI components intended to **track or manage student attendance.**
Profile	Displays the **teacher's personal information** and account details.

IX. IMPLEMENTATION

The mobile application implemented using **Jetpack Compose**, a modern declarative framework for Android UI development. This approach allows the interface to be written entirely in Kotlin, eliminating the need for XML layouts and enabling reactive and modular UI components.

Key Implementation Details:

1. **Screen Structure**
   - Each module—**Login, Dashboard, Schedule, Grades, Attendance, Profile**—is implemented as a separate composable function.
   - Screens are organized under **ui/screens/** for modularity, while reusable components such as buttons, cards, and toolbars are placed in **ui/components/.**
   - This structure ensures maintainability and allows easy addition of new modules in the future.

2. **Navigation System**
   - Navigation between screens is managed using **NavController** and **NavHost** from Android Jetpack Navigation.
   - The **Login Screen** is the start destination, and the **Dashboard** acts as the central hub for accessing all other modules.
   - Proper back stack handling ensures smooth transitions and a predictable navigation **flow**.
3. **UI Design and Principles**
   - The application follows **Material Design 3 guidelines**, emphasizing clarity, consistency, and accessibility.
   - Elements such as elevated cards, rounded buttons, and dynamic typography enhance readability and user experience.
   - The **Dashboard** provides a clear layout where teachers can access all modules efficiently.
4. **State Management and Reactivity**
   - Composables are reactive to UI state changes using **State** and **ViewModel**, allowing dynamic updates for components such as schedule and grades lists (mocked data for the prototype).

X. CONCLUSION

The **Teacher-Side Application** successfully demonstrates the design and development of a mobile user interface prototype intended for teachers of Colegio De Alicia. The project focuses on creating a clean and organized interface that allows teachers to navigate important modules such as the dashboard, schedule, grades, attendance, and profile.

The application was developed using **Android Studio**, **Kotlin**, and **Jetpack Compose**, which enabled the implementation of modern Android UI design practices. Through the use of composable functions and navigation components, the project demonstrates how a mobile interface can be structured to improve usability and organization for teachers.

Although the project is limited to the **UI/UX aspect of the system**, it provides a strong foundation for future development. With the addition of backend integration, database connectivity, and authentication systems, the application could be expanded into a fully functional teacher management platform that supports academic activities more efficiently.